VICTORY 2027

LIGHT

LECTURE-06 PHYSICS

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Topics To Be Covered

1. REVISION
2. COLOURS IN WHITE LIGHT WITH THEIR WAVELENGTH AND FREQUENCY RANGE
3. ELECTROMAGNETIC SPECTRUM
4. QUESTIONS


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REVISION

Electromagnetic Spectrum

→ Light Wave

SSS

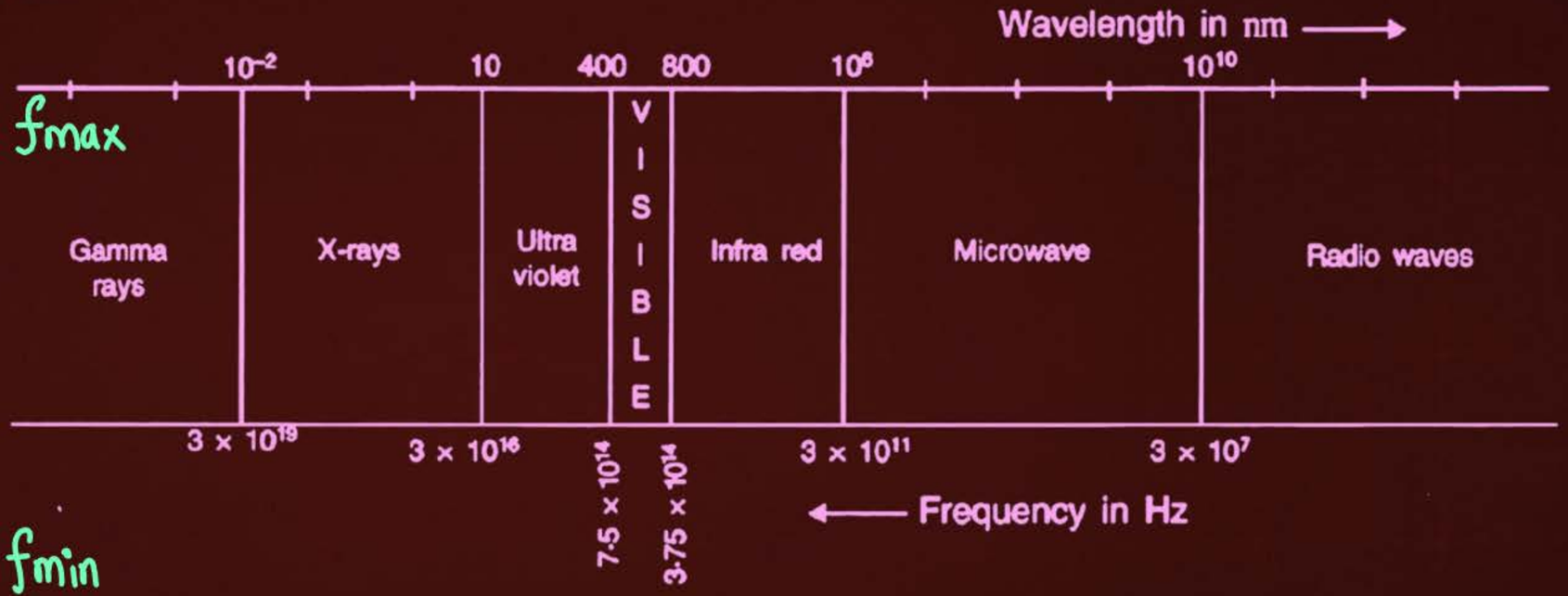
Gaadi X U V In My Range.

- Gaadi → Gamma
- X → X-Rays
- U → Ultraviolet
- V → Visible
- In → Infrared
- My → Microwaves
- Range → Radio

ELECTROMAGNETIC SPECTRUM

The complete electromagnetic spectrum in the increasing order of their wavelengths (or decreasing order of their frequencies) is given below :

(1) Gamma rays, → λ_{min} f_{max}
(2) X-rays,
(3) Ultraviolet rays,
(4) Visible light,
(5) Infrared radiations,
(6) Microwaves, and
(7) Radio waves → λ_{max} f_{min}



These electromagnetic waves are of different wavelengths, but they have properties similar to the light waves. Each wave travels with same speed equal to $3 \times 10^8 \text{ms}^{-1}$ in vacuum (or air).

PROPERTIES COMMON TO ALL THE ELECTROMAGNETIC WAVES

1. **No material medium is required** for their propagation. (Non-Mechanical Wave)
2. **All electromagnetic waves travel at the same speed in a given medium** (in vacuum/air: **3×10^8 m s^{-1}**).
3. They undergo **reflection and refraction**. During refraction, **speed, wavelength, and direction change, but frequency remains constant**.
4. They are **not deflected by electric or magnetic fields**.
5. They are **transverse waves**.

PROPERTIES AND USES OF DIFFERENT RADIATIONS OF ELECTROMAGNETIC SPECTRUM

Gamma Rays

- **Wavelength**: Less than **0.1 Å (0.01 nm)**; most energetic electromagnetic waves.
- **Sources**: Emitted by **radioactive nuclei** and found in **cosmic rays**.
- **Properties**: Cause **fluorescence**, have **very high penetrating power** (can pass through thick metal sheets), and can **damage living tissues**.
- **Uses**: Used in **cancer treatment (radiotherapy)** and **industrial inspection of welds**.

PROPERTIES AND USES OF DIFFERENT RADIATIONS OF ELECTROMAGNETIC SPECTRUM

X-Rays

- **Wavelength**: **0.1 Å – 100 Å (0.01–10 nm)**.
- **Source**: Produced when **high-speed cathode rays strike a heavy metal target**.
- **Properties**: Chemically more active than UV rays, **affect photographic plates**, **cause fluorescence**, penetrate **soft tissues** but are **absorbed by bones**.
- **Uses**: Used in **radiography** (bones and teeth), **CT/CAT scans**, **crystal structure analysis**, and **security screening** to detect concealed metals.

PROPERTIES AND USES OF DIFFERENT RADIATIONS OF ELECTROMAGNETIC SPECTRUM

Ultraviolet (UV) Radiations

- **Wavelength: 100 Å – 4000 Å (10–400 nm).**
- **Discovery:** First detected by **J. Ritter (1801)**.
- **Detection:**
  - Identified by their **strong chemical effect** (e.g., darkening of silver chloride).
  - Detected using **photographic plates** and **chemical dyes**.
  - Their spectrum is obtained using a **quartz prism**, as **glass absorbs UV rays**.

PROPERTIES AND USES OF DIFFERENT RADIATIONS OF ELECTROMAGNETIC SPECTRUM

Ultraviolet (UV) Radiations

Sources

- **Electric arcs and sparks**
- **Mercury vapour lamps**
- **The Sun** (most harmful UV rays are absorbed by the **ozone layer**)

Properties

- Pass through **quartz** but are **absorbed by glass**.
- Travel at **3×10^8 m s^{-1}** in vacuum (or air).
- Are **scattered by atmospheric dust**.
- Obey the **laws of reflection and refraction**.
- Have **strong chemical action**, affecting photographic plates.
- Produce **fluorescence** on materials like **zinc sulphide**.

PROPERTIES AND USES OF DIFFERENT RADIATIONS OF ELECTROMAGNETIC SPECTRUM

Harmful effect of ultraviolet radiations

The ultraviolet radiations cause health hazards like skin cancer if human body is exposed to them for a long period.

Uses of ultraviolet radiations

Ultraviolet radiations are used for the following purposes:

(a) For sterilising purposes.
(b) For detecting the purity of gems, eggs, ghee, etc.
(c) In producing vitamin D in food of plants and animals.

PROPERTIES AND USES OF DIFFERENT RADIATIONS OF ELECTROMAGNETIC SPECTRUM

Visible Light

- **Wavelength: 4000 Å – 8000 Å (400–800 nm)**.
- **Discovery:** Visible spectrum was discovered by **Newton** using a prism.
- **Colours: Violet, Indigo, Blue, Green, Yellow, Orange, Red (VIBGYOR)**.
- **Sources: Sun, electric bulbs, flames, and white-hot objects**.
- **Uses:** Used for **vision**, **photography**, and **photosynthesis**.

PROPERTIES AND USES OF DIFFERENT RADIATIONS OF ELECTROMAGNETIC SPECTRUM

Infrared (IR) Radiations

- **Wavelength: 8000 Å – 10^7 Å (800 nm – 1 mm).**
- **Discovery:** First detected by **William Herschel (1800).**
- **Detection:**
  - Identified by their **strong heating effect** beyond the red end of the visible spectrum.
  - Detected using a **blackened thermometer** or a **thermopile**.
  - Their spectrum is obtained using a **rock-salt prism**, as **glass absorbs infrared rays**.

PROPERTIES AND USES OF DIFFERENT RADIATIONS OF ELECTROMAGNETIC SPECTRUM

Infrared (IR) Radiations

Sources

- **Red-hot bodies** (heated iron, flames, fire, etc.)
- **The Sun** (natural source)

Properties

- Travel in straight lines at **3×10^8 m s^{-1}** in vacuum and obey **reflection and refraction**.
- Produce a **strong heating effect** and can be **focused by lenses and mirrors**.
- Do **not affect ordinary photographic films**, but affect **special IR-sensitive films**.
- Are **absorbed by glass** but **pass through rock salt**.
- Detected using a **blackened thermometer** or **thermopile**.
- Are **less scattered by the atmosphere** due to their **long wavelength**.

PROPERTIES AND USES OF DIFFERENT RADIATIONS OF ELECTROMAGNETIC SPECTRUM

Infrared (IR) Radiations

Harmful Effect

High exposure to infrared radiations can cause **skin burns**.

Uses

- Used in **heat therapy** (medical treatment).
- Used for **night photography** and imaging through **fog and mist**. (Haze Photography)
- **Infrared lamps** are used in **darkrooms** to develop photographs.
- Used for **military signalling**, as they are invisible and penetrate the atmosphere well.
- Used in **TV and electronic remote controls**.

PROPERTIES AND USES OF DIFFERENT RADIATIONS OF ELECTROMAGNETIC SPECTRUM

Microwaves

Wavelength **10^7 Å – 10^{11} Å (1 mm – 10 m).**

Source Produced by **electronic devices** such as **crystal oscillators**.

Uses:

- **Satellite communication**
- **Radar systems**
- **Microwave ovens** (cooking)
- **Study of atomic and molecular structure**

PROPERTIES AND USES OF DIFFERENT RADIATIONS OF ELECTROMAGNETIC SPECTRUM

Radio Waves

- **Wavelength:** **Greater than 10 m (10^{11} Å)**; **longest wavelength** in the electromagnetic spectrum.
- **Frequency:** **Less than 3×10^7 Hz.**
- **Properties:** Exhibit all the **properties of electromagnetic waves**.
- **Uses:** Used in **radio and television broadcasting** and **radar communication**.

Comparison of Ultraviolet, Visible and Infrared Radiations

HW

Property	Ultraviolet (UV) Radiations	Visible Radiations	Infrared (IR) Radiations
Wavelength Range	100 Å – 4000 Å	4000 Å – 8000 Å	8000 Å – 10^7 Å
Visibility	Invisible	Visible	Invisible
Heating Effect	No heating effect	Slight heating effect	Strong heating effect
Effect on Photographic Plate	Affects photographic plate	Affects photographic plate	Does not affect ordinary photographic plate
Fluorescence	Causes fluorescence (e.g., zinc sulphide)	Does not cause fluorescence	Does not cause fluorescence
Effect on Human Body	May cause health hazards such as skin cancer	No harmful effect under normal conditions	High doses may cause skin burns
Transmission Through Materials	Passes through quartz, not glass	Passes through glass	Passes through rock salt, not glass

QUESTIONS

The most energetic electromagnetic radiations are:

1. microwaves
2. ultraviolet waves
3. X-rays
4. gamma rays

QUESTIONS

The source of ultraviolet light is:

1. electric bulb
2. red hot iron ball
3. sodium vapour lamp
4. carbon arc-lamp

QUESTIONS

A radiation P is focused by a proper device on the bulb of a thermometer. Mercury in the thermometer shows a rapid increase. The radiation P is:

1. infrared radiation
2. visible light
3. ultraviolet radiation
4. X-rays

SCATTERING OF LIGHT

- **Scattering** is the **absorption and re-emission of light** by tiny particles such as **dust and air molecules**.
- When sunlight enters Earth's atmosphere, it is **scattered in all directions**, making the sky appear bright.
- The **path of sunlight** entering a dark room through a small hole becomes visible due to **scattering by dust particles**.
- **Rayleigh** was the first scientist to study the **scattering of light**.



SCATTERING OF LIGHT

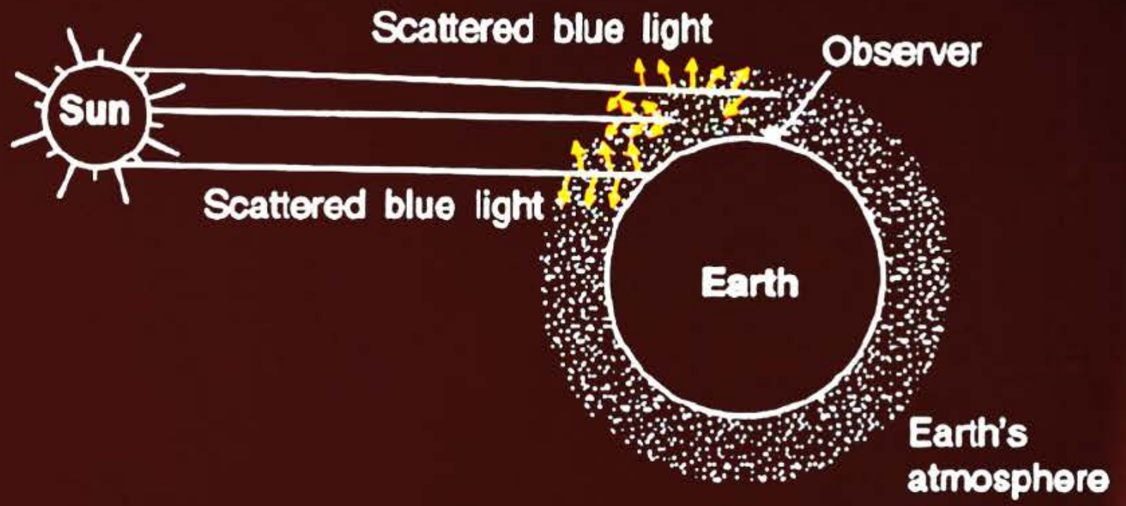
Rayleigh Scattering

- **Air molecules smaller than the wavelength of light** absorb and re-emit light **without changing its wavelength**.
- The **intensity of scattered light** is inversely proportional to the **fourth power of wavelength**: $\text{Scattering} \propto \frac{1}{\lambda^4}$
- Hence, **shorter wavelengths scatter more** than longer wavelengths.
- **Violet light (≈4000 Å)** is scattered the **most**, while **red light (≈8000 Å)** is scattered the **least**.
- **Violet light is scattered about 16 times more than red light.**
- Therefore, sunlight reaching Earth's surface contains **more red/orange light** and **less violet/blue light**.
- **Larger particles** (size greater than the wavelength of light) **scatter all wavelengths almost equally**.

SOME APPLICATIONS OF SCATTERING

(1)RED COLOUR OF SUN AT SUNRISE AND SUNSET

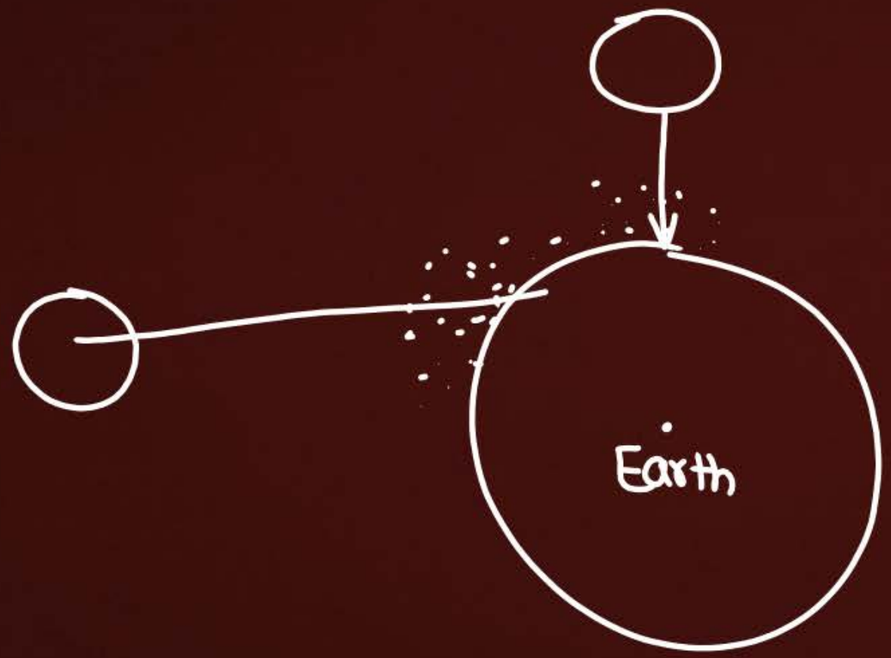
- At **sunrise and sunset**, sunlight travels the **maximum distance through the atmosphere**.
- **Blue light** (short wavelength) is **scattered the most**, so most of it is removed from the direct sunlight.
- **Red light** (long wavelength) is **scattered the least** and reaches the observer.
- Hence, the **Sun and the surrounding sky appear reddish** at sunrise and sunset.



SOME APPLICATIONS OF SCATTERING

(2) WHITE COLOUR OF SKY AT NOON

At noon, sun is directly above our head, so we get the light rays directly from sun after travelling the shortest distance, without much scattering of any particular colour. Hence the sky is seen white.



SOME APPLICATIONS OF SCATTERING

- **Why is the Sky Blue?**

- Sunlight is **scattered by air molecules** as it passes through Earth's atmosphere.
- **Blue (and violet) light**, having a **shorter wavelength**, is **scattered much more** than red light.
- The **scattered blue light** reaches our eyes from all directions, making the **sky appear blue**.
- The **direct sunlight** reaching us is comparatively **richer in red light** because less red light is scattered.
- **After sunset and before sunrise**, only the **scattered blue light** is visible, so the **sky continues to appear blue**.

SOME APPLICATIONS OF SCATTERING

Why is the Sky Black in the Absence of Atmosphere?

- **Without an atmosphere, there is no scattering of sunlight**, so the **sky appears black**.
- The **Moon has no atmosphere**, hence its sky looks **black** except in the direction of the Sun.
- Similarly, an **astronaut above Earth's atmosphere** sees the **sky as black**, while the **Earth appears blue** due to scattering by Earth's atmosphere.
- **Stars do not twinkle** in space because there is **no atmospheric interference**.

SOME APPLICATIONS OF SCATTERING

HW

Why are Clouds White?

- Clouds contain **dust particles and tiny water droplets/ice crystals** that are **larger than the wavelength of visible light**.
- These particles **scatter all colours of sunlight equally**.
- The combination of all scattered colours appears **white** to our eyes.
- Hence, **clouds appear white**.

SOME APPLICATIONS OF SCATTERING

Why is Red Light Used for Danger Signals?

- **Red light has the longest wavelength** in the visible spectrum.
- It is **scattered the least** by air molecules, so it **travels the farthest** without losing much intensity.
- Therefore, **red signals remain visible from long distances**, even in **fog or dust**.
- Hence, **red light is used for danger and stop signals**.
- **Infrared radiations** are also used in **remote controls** for electronic devices such as TVs.

QUESTIONS

In the white light of sun, maximum scattering by the air molecules present in the earth's atmosphere is for:

1. red colour
2. yellow colour
3. green colour
4. blue colour

QUESTIONS

To an astronaut in a space-ship, the earth appears:

1. white
2. red
3. blue
4. black

ELECTROMAGNETIC SPECTRUM

HW

Name of the Wave	Wavelength (nm)	Discoverer	Source	Method of Detection
Gamma rays	Below 0.01	Becquerel & Curie	In cosmic rays; from radioactive substances	By their large penetrating power
X-rays	0.01 – 1.0	Roentgen	From a heavy metal target of high melting point when highly energetic electrons are stopped by it	By the fluorescence produced on a zinc sulphide screen. The photographic film gets affected.
Ultraviolet	1.0 – 400	Ritter	Sunlight, arc-lamp, spark	By their chemical activity on dyes. Photographic plates get affected. It causes fluorescence.

ELECTROMAGNETIC SPECTRUM

Name of the Wave	Wavelength (nm)	Discoverer	Source	Method of Detection
Visible Light	400 – 800	Newton	Sunlight, light from electric bulb, flame, white hot bodies	Other objects can be seen in its presence
Infrared Waves	800 – 10^6	Herschel	Lamp with thoriated filament, heated silicon carbide rod, red hot bodies	Heating effect is more. The mercury rises rapidly when a thermometer with a blackened bulb is kept in these radiations.
Microwaves	10^6 – 10^{10}	Hertz	Electronic devices such as crystal oscillators	Oscillatory electrical circuit.
Radio Waves	Above 10^{10}	Marconi	TV and radio transmitters	Aerials of radio and TV receiver.

APPROXIMATE RANGES OF WAVELENGTH AND FREQUENCY

HW

Region of Electromagnetic Spectrum	Wavelength (Å)	Wavelength (SI Units)	Frequency Range (Hz)
▪ Gamma Rays	< 0.1 Å	$\lambda < 0.01$ nm	$> 10^{19}$ Hz
▪ X-Rays	0.1 Å – 100 Å	0.01 nm – 10 nm	$3 \times 10^{19} – 3 \times 10^{16}$ Hz
▪ Ultraviolet (UV) Rays	100 Å – 4000 Å	10 nm – 400 nm	$3 \times 10^{16} – 7.5 \times 10^{14}$ Hz
▪ Visible Light	4000 Å – 8000 Å	400 nm – 800 nm	$7.5 \times 10^{14} – 3.75 \times 10^{14}$ Hz
▪ Infrared (IR) Radiations	8000 Å – 10^7 Å	800 nm – 1 mm	$3.75 \times 10^{14} – 3 \times 10^{11}$ Hz
▪ Microwaves	10^7 Å – 10^{11} Å	1 mm – 10 m	$3 \times 10^{11} – 3 \times 10^7$ Hz
▪ Radio Waves	$> 10^{11}$ Å	$\lambda > 10$ m	$< 3 \times 10^7$ Hz

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.

Thank You